SANJAY P

Machine Learning Engineer | Computer Vision | Autonomous Systems

+91-7339514516 ◇ Chennai, India ◇ sanjayp400080@gmail.com

SUMMARY

Highly motivated Computer and Communication Engineering student specializing in **Deep Learning, Computer Vision, and Autonomous Navigation**. Experienced in architecting end-to-end AI pipelines—from processing 3D LiDAR point clouds to deploying real-time inference applications. Proven ability to simulate complex environments in CARLA and optimize machine learning models for high-density traffic scenarios.

EDUCATION

Bachelor of Technology in Computer and Communication Engineering Expected May 2027
Amrita School of Engineering, Chennai, India
Relevant Coursework: Deep Learning, Probability & Statistics, Data Structures & Algorithms, Linear Algebra, Object-Oriented Programming, Digital Signal Processing.

TECHNICAL SKILLS

- **AI/ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, Keras, Hugging Face.
- **Computer Vision:** OpenCV, YOLO (v8/v10), Image Segmentation, MediaPipe.
- **Data Science & Tools:** Pandas, NumPy, LiDAR Point Cloud Processing, Matplotlib, Seaborn, Streamlit.
- **Programming & Databases:** Python (Expert), C++, SQL (PostgreSQL/MySQL), Git/GitHub.
- **Domains:** Autonomous Driving (CARLA), ADAS Prototyping, Embedded AI, Sensor Fusion.

MACHINE LEARNING & AI PROJECTS

Contextual AI for ADAS (CARLA Simulator)
- Developed a **situational-aware perception system** for unstructured road environments using multi-sensor fusion.
- Implemented **YOLO-based object detection** to handle dense traffic, improving detection accuracy in occluded scenarios.
- Integrated real-time collision avoidance logic by processing environmental data within the CARLA Python API.

LiDAR-Based Perception & Emergency Braking
- Engineered a 3D environment perception pipeline using **LiDAR point cloud data** for obstacle localization.
- Developed an **Automatic Emergency Braking (AEB)** algorithm, reducing simulated collision risk by optimizing the braking threshold based on distance-to-impact calculations.
- Utilized Open3D for visualizing and preprocessing raw LiDAR data for downstream navigation tasks.

Livestock Breed Classification (Smart India Hackathon)
- Architected a **Convolutional Neural Network (CNN)** to classify cow and buffalo breeds with high precision.
- Leveraged **Transfer Learning (ResNet/EfficientNet)** and data augmentation techniques to mitigate class imbalance.
- Performed extensive hyperparameter tuning to optimize model weights for deployment on low-power edge devices.

Predictive Analytics for Vehicle Platooning
- Applied **Regression and Time-Series analysis** to vehicle telemetry data to optimize fuel efficiency in convoys.
- Modeled the correlation between inter-vehicle spacing and aerodynamic drag using simulated sensor logs.

Deployment-Ready ML Web Applications
- Built and deployed interactive ML dashboards using **Streamlit**, enabling real-time model testing and visualization.
- Managed database integration with SQL for logging inference results and metadata for model monitoring.

ACHIEVEMENTS AND LEADERSHIP

- **2nd Place (Winner):** Roba Maze Robotics Competition at Tantrostav Tech Fest; optimized maze-solving algorithms using autonomous navigation logic.
- **Media Co-Head, CAMHI Club:** Led a team of 10+ members to coordinate technical event publicity and digital branding for AI-focused workshops.